



COVENANT UNIVERSITY
CANAAANLAND, KM 10, IDIROKO ROAD
P.M.B 1023, OTA, OGUN STATE, NIGERIA
B.Sc. DEGREE EXAMINATION

COVENANT UNIVERSITY OTA
COMPUTER & INFO. SCIENCES

HEAD OF DEPARTMENT

Date: 25/2/24

COLLEGE: Science and Technology
SESSION: 2023/2024
COURSE CODE: CSC213
COURSE TITLE: Structured Programming
INSTRUCTION: Answer any FOUR (4) Questions

DEPARTMENT: Computer & Info. Sciences
SEMESTER: ALPHA
CREDIT UNIT: 3
TIME: 3 Hours

Question One

- a. Briefly describe how the *and* operator works in Python (use examples) [2.5 Marks]
b. Given numbers = [1, 2, 3, 4, 5]. What will be the output of numbers*2 [3 Marks]
c. Identify the Errors in the following code segment [4 Marks]

```
1 person = { 'name': 'John Doe', 'age': 30, 'city':  
2 'New York', 'Email': 'john.doe@example.com' }  
3  
4 print("Name:", person['Name'])  
5 print("Age:", person['Age'])  
6 print("City:", person['City'])  
7 print("Email:", person['email'])  
8  
9 person['Age'] = "35"  
10 person['phone'] = '123-456-7890'  
11 del person['email']  
12  
13 print("Updated Person Dictionary:")  
14 print(person)  
15
```

- d. In Nigeria, property taxes are calculated based on the assessment value of the property, which is typically 60 percent of its actual value. For instance, if a plot of land is valued at ₦100,000, its assessment value would be ₦60,000. The property tax rate varies depending on the assessed value. For properties with an assessment value of ₦10,000 or less, the tax rate is ₦0.72 for each ₦100 of the assessment value. However, for properties with an assessment value exceeding ₦10,000, the tax rate increases to ₦0.85 for each ₦100 of the assessment value. Write a Python program that prompts the user to input the actual value of a property and then calculates and displays the corresponding assessment value along with the property tax based on the applicable tax rates [Hint: Use Functions]. [8 Marks]

QUESTION TWO

- a. What is a count-controlled loop in Python? [2.5 Marks]
b. Identify the Errors in this code segment and write the corrected version. [3 Marks]

```

1 my_string = "Serve Jesus Now"
2 print("Original String:", mystring)
3
4 first_character = my_string[1]
5 last_character = my_string[-1]
6 print("First Character:", first_character)
7 print("Last Character:" last_character)
8
9 substring = my_string[17:]
10 print("Substring:", substring)
11
12 greeting = "Hi"
13 name = "Alice"
14 message = greeting + ", " + name + "!"
15 print("Message:", message)
16
17 length = len(mystring)
18 print("Length of the String:", length)
19
20 count_e = my_string.count('e')
21 print("Count of 'e':", count_e)
22

```

- c. Write a Python program that generates 100 random numbers between 1 and 10. The program should then store the frequency of each number generated in a dictionary, where the number is the key and the frequency is the value. For example, if the program generates the number 6 a total of 11 times, the dictionary will contain a key of 6 with a value of 11. After storing the frequencies, the program should display information about the frequency of each number. Additionally, the program should calculate and display the sum of all the 100 random numbers generated using the formula $\text{Key} \times \text{Value}$.

[12 Marks]

Question Three

- a. Mention five (5) benefits of modularizing a program with functions [2.5 Marks]
 b. Look at the following statement: levels = "Beginner, Average, Advanced, Expert". Write a statement that splits this string, creating a list of the competence levels. [2 Marks]
 c. What is are the outputs given the following code segment [3 Marks]

```

1
2 nested_list = [
3     [1, 2, 3],
4     [4, 5, 6],
5     [7, 8, 9],
6     [10, 11, 12, 13],
7     [14, [15, 16, 17], 18],
8     [19, [20, [21, 22], 23], 24]
9 ]
10

```

- i. What is the value at index [3][2]?
 ii. print (len (nested_list[4]))
 iii. What is the value at index [5][1][1][0]?

- d. Suppose there is a file named `financial_data.txt` stored on a computer's disk. This file contains financial records, each consisting of two fields: a company name followed by its yearly revenue (expressed in millions of Naira). Write a Python program to analyze the data in `financial_data.txt` and display the company with the highest revenue, along with its revenue value. Additionally, your program should report the total number of companies included in the file. [Hint: Utilize a variable and an "if" statement to track the highest revenue found while iterating through the records. Also, maintain a variable to count the total number of records in the file.] [10 Marks]

Question Four

- a. Distinguish between a Void and Value-returning function in Python (with examples). [2.5 Marks]
b. What will `list1` and `list2` contain after the following statements are executed?

i. `list1 = [1, 2] * 2`

ii. `list2 = [3]`

iii. `list2 += list1`

[2 Marks]

- c. Identify the Error in this code Segment

[3 Marks]

```
1
2 with open(example.txt, 'w') as file:
3     file.write("Hello, World!\n")
4     file.write("This is a sample file.\n")
5     file.write("It contains some text.\n")
6
7 # Reading data from the file
8 with open(example.txt, 'r') as file:
9     # Read the entire content of the file
10    content = file.reading()
11
12 # Print the content read from the file
13 print(content)
14
```

- d. Imagine you're using an application that converts your favorite 9-letter word into a unique numeric format. This application employs a special mapping where alphabetic characters correspond to numbers as shown below. Write a Python program that interacts with users, prompting them to input their favorite 9-letter word. Once entered, the program should convert the word into its unique numeric format based on the provided mapping and display the result. For example, if the user enters "HIGHLIGHT", the program should output the corresponding numeric format which is 444454448. [10 Marks]

A, B, and C = 2

D, E, and F = 3

G, H, and I = 4

J, K, and L = 5

M, N, and O = 6

P, Q, R, and S = 7

T, U, and V = 8

W, X, Y, and Z = 9

Question Five

a. Highlight the three OS File Modes in Python

[3 Marks]

b. A program contains the following function definition:

[2 Marks]

```
2
3 - def cube(num):
4     return num * num * num
5
```

- i. Write a statement that passes the value 4 to this function and assigns its return value to the variable result.
 - ii. What will be the value in result
- c. Design a Python program that interacts with users, prompting them to enter a series of 20 numbers. The program should store these numbers in a list and then display the following data:
- i. The lowest number in the list.
 - ii. The highest number in the list.
 - iii. The total sum of the numbers in the list.
 - iv. After displaying the lowest, highest, and total sum of the numbers, the program should offer users the following choice to calculate the average of the numbers in the list Including or excluding the highest and lowest numbers.
 - v. Note that if any of the entered numbers are negative, prompt the user to enter them again until all 20 numbers are non-negative. If after five attempts a non-negative number is not provided, terminate the program with an appropriate message.

[12.5 Marks]

Question Six

a. Strings are *Immutable*. Explain and give reason why.

[2.5 Marks]

b. Write a function that accepts a string as an argument and returns true if the argument starts with the substring 'https'. Otherwise, the function should return false.

[8 Marks]

c. Given three Bible scriptures stored as strings:

Str1= "For God so loved the world that he gave his one and only Son, that whoever believes in him shall not perish but have eternal life."

Str2 = "The Lord is my shepherd, I lack nothing."

Str3 = "For I know the plans I have for you, declares the Lord, plans to prosper you and not to harm you, plans to give you hope and a future."

Write a Python program that performs the following Set operations on the words in these scriptures:

[12 Marks]

- i. Create a set containing all unique words found in the scriptures.
- ii. Display the set of unique words from the scriptures.
- iii. Calculate the intersection of the sets formed by the words in each scripture.
- iv. Calculate the union of the sets formed by the words in each scripture.
- v. Calculate the difference between the sets formed by the words in each scripture.